

YEAR 7 MATHEMATICS · CHAPTER 8

Area and volume

- | | |
|--|--------------------------------------|
| 8.01 Units of measurement | 8.02 Perimeter |
| 8.03 Circumference of a circle | 8.04 Area units |
| 8.05 Area of squares and rectangles | 8.06 Area of a triangle |
| 8.07 Area of a parallelogram | 8.08 Area of composite shapes |
| 8.09 Prisms, cross-sections and views | 8.10 Volume units |
| 8.11 Volume of a rectangular prism | 8.12 Volume of a prism |
| 8.13 Capacity | |

Name _____

Class _____

Teacher _____

How to use this booklet

- Each lesson starts with **learning intentions** and **success criteria**. Write the **key terms** as your teacher explains them, and use the **notes** box.
- Work through the levels in order: Level 1 , then Level 2, then Level 3. Extension questions stretch you further.
- Some questions ask you to **draw** a diagram, graph or construction. Use the space given, a ruler and a sharp pencil.
- At the end of each lesson, **show off your skill**: choose one question, then tear off the ticket and hand it in.
- Missed a lesson? Each lesson ends with a **summary** that has a worked example for every level. Start there, then try the questions.

WE DO

Grey boxes with a dark border. Work with your teacher. Copy what they write on the board.

YOU DO

Blue boxes. Work on your own. Show your working in the box and write your answer.

Contents

Lesson	Topic	Page	Done
8.01	Units of measurement	3	☐
8.02	Perimeter	7	☐
8.03	Circumference of a circle	11	☐
8.04	Area units	15	☐
8.05	Area of squares and rectangles	19	☐
8.06	Area of a triangle	23	☐
8.07	Area of a parallelogram	27	☐
8.08	Area of composite shapes	31	☐
8.09	Prisms, cross-sections and views	35	☐
8.10	Volume units	39	☐
8.11	Volume of a rectangular prism	43	☐
8.12	Volume of a prism	47	☐
8.13	Capacity	51	☐

In this chapter you will

- convert units of length, mass, capacity and time
- find perimeter and circumference
- convert area units and find the area of rectangles, triangles, parallelograms and composite shapes
- draw cross-sections and views of prisms
- convert volume units and find the volume of prisms
- find capacity in litres

NSW Mathematics K-10 Syllabus (2022), Stage 4 Measurement and space: MA4-LEN-C-01 (length and circumference), MA4-ARE-C-01 (area) and MA4-VOL-C-01 (volume and capacity).



Acknowledgement of artwork

The artwork used throughout this booklet series was created by **Marni Tuala**. We thank Marni for sharing this artwork with our school community.

Acknowledgement of Country

We acknowledge the Bundjalung people, the Traditional Custodians of the land on which Kingscliff High School stands, and pay our respects to Elders past and present.

LESSON
8.01

Units of measurement

Date _____

Learning intentions

We are learning to:

- convert between units of length, mass, capacity and time

Success criteria

I can:

- ☐ know the metric prefixes: kilo, centi, milli
- ☐ multiply to convert to a smaller unit, divide to convert to a larger unit
- ☐ convert units of time

Key terms Write the definition as your teacher explains it.**Metric****Convert**

Notes

Level 1 · Getting started

WE DO With your teacher: copy what they write on the board.**Example 1** Convert 3200 m to km.**Example 2** Convert 0.042 kg to g.**YOU DO** On your own: Set A. Convert.

1 3200 m to km

2 904 cm to mm

3 7150 mm to m

4 0.042 kg to g

5 3.5 kg to t

6 125 L to kL

YOU DO On your own: Set B. Convert.

1 8.3 L to mL

4 15 h to min

2 4.2 kL to mL

5 2.5 days to h

3 240 min to h

6 17 000 000 mg to kg

Level 2 · Building up**WE DO** With your teacher: copy what they write on the board.**Example 3** Convert 8.3 L to mL.**Example 4** Convert 240 min to h.**YOU DO** On your own: Set C. Convert. Show your working.

1 15 h to s

3 4500 g to kg

5 750 mL to L

2 0.65 km to cm

4 3 h 20 min to min

6 1 week to min



Level 3 · Challenge

Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.**Example 5** Convert 15 h to seconds.
Example 6 How many minutes has Isla lived by her 12th birthday? (365 days a year)
YOU DO On your own: Set D. Problems.

- 1 How many minutes has Isla lived by her 12th birthday? (365 days a year)

- 3 A tap drips 5 mL a minute. How many litres in one day?

- 2 A recipe needs 350 g of flour for one cake. How many kg for 12 cakes?

- 4 Order from smallest to largest: 1.2 km, 950 m, 125 000 cm, 1 100 000 mm.

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO Show off your skillChoose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Convert 4.5 m to cm.

Level 2

Convert 2750 g to kg.

Level 3

Convert 3.5 h to minutes.

Extension · Stretch yourself

Worked example

Light travels 300 000 km in one second. How far does it travel in one hour?

1. 1 hour = 3600 s
2. $300\,000 \times 3600$

Answer: 1 080 000 000 km

YOU DO

On your own: read the worked example, then try these.

- E1** Your heart beats about 70 times a minute. About how many times does it beat in a year?

- E2** Convert 1 million seconds to days (1 d.p.).

Lesson summary

Missed the lesson? Start here.

Conversions 1. $\text{km} \rightarrow \text{m} \times 1000$, $\text{m} \rightarrow \text{cm} \times 100$, $\text{cm} \rightarrow \text{mm} \times 10$. 2. $\text{kg} \rightarrow \text{g} \times 1000$, $\text{t} \rightarrow \text{kg} \times 1000$, $\text{L} \rightarrow \text{mL} \times 1000$.
3. Bigger unit $\rightarrow$ smaller unit: multiply. Smaller $\rightarrow$ bigger: divide. 4. Time: $60 \text{ s} = 1 \text{ min}$, $60 \text{ min} = 1 \text{ h}$, $24 \text{ h} = 1 \text{ day}$.

Level 1

3200 m to km

1. $\div 1000$

Answer: 3.2 km

Level 2

8.3 L to mL

1. $\times 1000$

Answer: 8300 mL

Level 3

15 h to s

1. $15 \times 60 \times 60$

Answer: 54 000 s

Extension

Light in 1 hour

1. $300\,000 \times 3600$

Answer: 1 080 000 000 km

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LESSON
8.02

Perimeter

Date _____

Learning intentions

We are learning to:

- find the perimeter of shapes

Success criteria

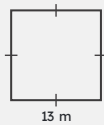
I can:

- ☐ add all the side lengths
- ☐ find missing side lengths first
- ☐ use the same units for every side

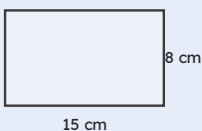
Key terms Write the definition as your teacher explains it.**Perimeter****Composite shape**

Notes

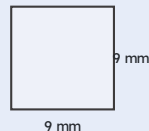
Level 1 · Getting started

WE DO With your teacher: copy what they write on the board.**Example 1** Find the perimeter of the square.**Example 2** Find the perimeter.**YOU DO** On your own: Set A. Find the perimeter.

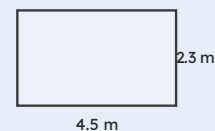
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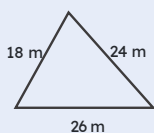
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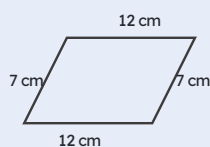
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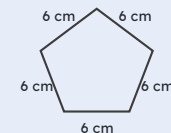
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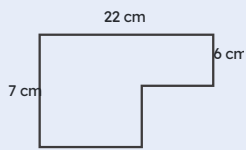


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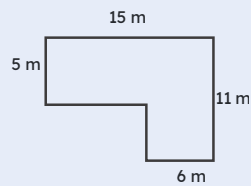


YOU DO On your own: Set B. Find the perimeter.

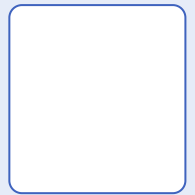
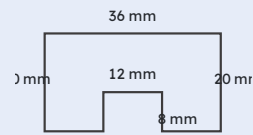
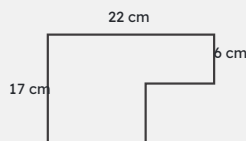
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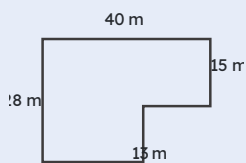
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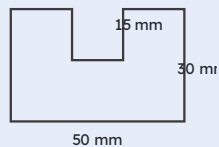
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**Level 2 · Building up****WE DO** With your teacher: copy what they write on the board.**Example 3** Find the perimeter.**Example 4** Find the perimeter of a rectangle 2.4 m by 85 cm (in cm).**YOU DO** On your own: Set C. Find the perimeter. Find missing sides first.

1

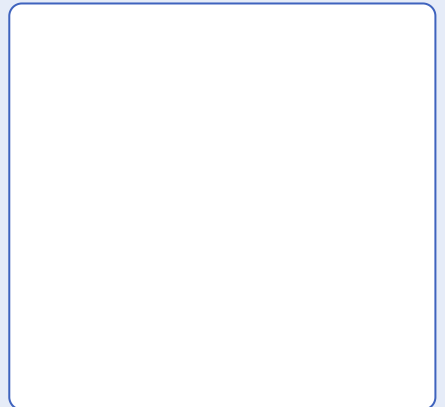


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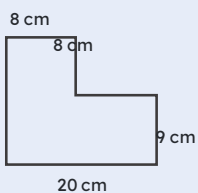


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A square has perimeter 46 cm. Find its side.



2



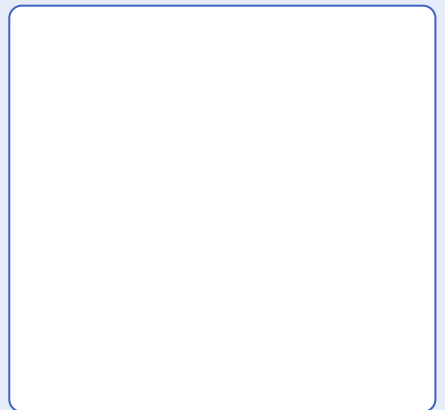
4

A rectangle is 2.4 m by 85 cm. Find its perimeter in cm.



6

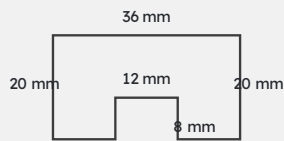
A rectangle has perimeter 30 m and length 9 m. Find its width.





Level 3 · Challenge

Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.**Example 5** Find the perimeter.**Example 6** A square has perimeter 46 cm. Find its side length.**YOU DO** On your own: Set D. Problems.

- 1 Fencing costs \$24.50 per metre. Find the cost to fence a 35 m by 18 m paddock.

- 3 How many different rectangles with whole-number sides have a perimeter of 20 cm?

- 2 A regular hexagon has perimeter 57 cm. Find the length of one side.

- 4 A photo 15 cm by 10 cm has a 2 cm frame around it. Find the outside perimeter of the frame.

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO Show off your skillChoose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Find the perimeter of a rectangle 12 cm by 7 cm.

Level 2

Find the perimeter of an equilateral triangle with sides 9.5 m.

Level 3

A square has perimeter 60 mm. Find its side.

 Extension · Stretch yourself

 Worked example

A rectangle's length is 3 times its width. Its perimeter is 64 cm. Find its dimensions.

1. Width w , length $3w$
2. $2(w + 3w) = 8w = 64$
3. $w = 8$

Answer: 8 cm by 24 cm

YOU DO

On your own: read the worked example, then try these.

- E1** A rectangle's length is 5 cm more than its width. Its perimeter is 50 cm. Find its dimensions.

- E2** An isosceles triangle has perimeter 40 cm and base 12 cm. Find the equal sides.

Lesson summary

Missed the lesson? Start here.

Remember

1. Perimeter = distance around the outside.
2. Add every side length.
3. Find missing sides using opposite sides.
4. Check all units are the same.

 Level 1

Square, side 13 m

1. 4×13

Answer: 52 m

 Level 2

Rectangle 13 mm $\times$ 5 mm

1. $2 \times (13 + 5)$

Answer: 36 mm

 Level 3

L-shape 22, 6, 17...

1. Missing sides: $22 - 10 = 12$, $17 - 6 = 11$

Answer: 78 cm

 Extension

Length = $3 \times$ width, $P = 64$

1. $8w = 64$

Answer: 8 cm by 24 cm

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LESSON
8.03

Circumference of a circle

Date _____

Learning intentions

We are learning to:

- find the circumference of a circle

Success criteria

I can:

- ☐ name the parts of a circle: centre, radius, diameter, circumference
- ☐ use $C = \pi d$ or $C = 2\pi r$
- ☐ round the answer to 1 decimal place

Key terms Write the definition as your teacher explains it.

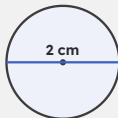
Circumference

Radius

Diameter

Notes

Level 1 · Getting started

WE DO With your teacher: copy what they write on the board.**Example 1** Find the circumference (1 d.p.).**Example 2** Find the circumference (1 d.p.).**YOU DO** On your own: Set A. Find the diameter of each circle.

1 radius 5 cm

4 radius 0.8 km

2 radius 12 m

5 radius 21 cm

3 radius 3.5 mm

6 radius 9 m

YOU DO On your own: Set B. Find the circumference, $C = \pi d$ (1 d.p.).

1 $d = 2 \text{ cm}$

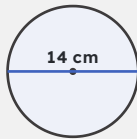
4 $d = 30 \text{ mm}$

2 $d = 14 \text{ cm}$

5 $d = 1.5 \text{ km}$

3 $d = 7 \text{ m}$

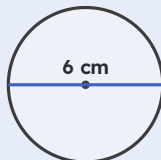
6 $d = 100 \text{ cm}$

Level 2 · Building up**WE DO** With your teacher: copy what they write on the board.**Example 3** Find C (1 d.p.).**Example 4** Find C (1 d.p.).**YOU DO** On your own: Set C. Find the circumference (1 d.p.).

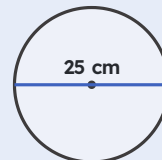
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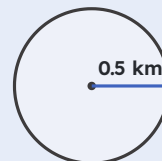
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4



6





Level 3 · Challenge

Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.**Example 5** A bike wheel has diameter 66 cm. How far does it travel in 100 turns (in m)?
Example 6 A circle has circumference 50 cm. Find its diameter (1 d.p.).
YOU DO On your own: Set D. Problems.**1** A bike wheel has diameter 66 cm. How far does it travel in 100 turns (in m, 1 d.p.)?
3 Find the perimeter of a semicircle with diameter 12 cm (1 d.p.).
2 A circle has circumference 50 cm. Find its diameter (1 d.p.).
4 A round table has radius 60 cm. How many 30 cm-wide chairs fit around it?

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO

Show off your skill

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

A circle has radius 8 cm. Find its diameter.

Level 2

Find the circumference of a circle with diameter 9 m (1 d.p.).

Level 3

Find the circumference of a circle with radius 3.5 cm (1 d.p.).

Extension · Stretch yourself

Worked example

Find the perimeter of a quarter circle with radius 10 cm (1 d.p.).

1. Arc = $\frac{1}{4} \times 2\pi \times 10 = 15.707\dots$
2. Add two radii: $15.71 + 20$

Answer: 35.7 cm

YOU DO

On your own: read the worked example, then try these.

- E1** Find the perimeter of a semicircle with radius 7 m (1 d.p.).

- E2** A running track has two straights of 100 m and two semicircles of diameter 64 m. Find its length (1 d.p.).

Lesson summary

Missed the lesson? Start here.

Remember

1. Diameter = $2 \times$ radius.
2. $C = \pi d$ or $C = 2\pi r$.
3. Use the π key on your calculator.
4. Round to 1 decimal place and include units.

Level 1

$d = 2$ cm

1. $\pi \times 2 = 6.283\dots$

Answer: 6.3 cm

Level 2

$r = 10$ mm

1. $2 \times \pi \times 10$

Answer: 62.8 mm

Level 3

$C = 50$, find d

1. $50 \div \pi$

Answer: 15.9 cm

Extension

Quarter circle $r = 10$

1. $15.7 + 20$

Answer: 35.7 cm

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LESSON
8.04

Area units

Date _____

Learning intentions

We are learning to:

- convert between units of area

Success criteria

I can:

- ☐ know $1 \text{ cm}^2 = 100 \text{ mm}^2$, $1 \text{ m}^2 = 10\,000 \text{ cm}^2$
- ☐ multiply to convert to a smaller unit, divide to a larger unit
- ☐ choose a sensible unit of area

Key terms Write the definition as your teacher explains it.

Area

Square unit

Notes

Level 1 · Getting started

WE DO With your teacher: copy what they write on the board.**Example 1** Why is $1 \text{ cm}^2 = 100 \text{ mm}^2$?**Example 2** How many cm^2 in 9.1 m^2 ?**YOU DO** On your own: Set A. Convert.1 3 cm^2 to mm^2

4 4 m^2 to cm^2

2 2.5 cm^2 to mm^2

5 2 km^2 to m^2

3 9.1 m^2 to cm^2

6 5 ha to m^2

YOU DO On your own: Set B. Convert.

1 240 mm^2 to cm^2

4 850 mm^2 to cm^2

2 $175\,000 \text{ cm}^2$ to m^2

5 $3\,500\,000 \text{ m}^2$ to km^2

3 $50\,000 \text{ m}^2$ to ha

6 6400 cm^2 to m^2

Level 2 · Building up**WE DO** With your teacher: copy what they write on the board.**Example 3** How many mm^2 in 2.5 cm^2 ?**Example 4** How many m^2 in $175\,000 \text{ cm}^2$?**YOU DO** On your own: Set C. Convert. Show your working.

1 $240\,000 \text{ mm}^2$ to m^2

3 0.07 km^2 to ha

5 0.35 m^2 to cm^2

2 8.6 m^2 to mm^2

4 12.5 ha to m^2

6 72 cm^2 to m^2



Level 3 · Challenge

Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.**Example 5** How many m^2 in 240 000 mm^2 ?
Example 6 How many mm^2 in 8.6 m^2 ?
YOU DO On your own: Set D. Choose and compare.

- 1 Which unit would you use for: a stamp, a classroom floor, a farm, a country?

- 3 A tile is 400 cm^2 . How many tiles cover 6 m^2 ?

- 2 Which is larger: 3.2 m^2 or 30 000 cm^2 ?

- 4 A farm is 2.4 km^2 . How many hectares is that?

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO Show off your skillChoose **ONE** question. Show your working.

Name _____ Class _____

Level 1Convert 5 cm^2 to mm^2 .
Level 2Convert 30 000 cm^2 to m^2 .
Level 3Convert 1.5 m^2 to mm^2 .

 Extension · Stretch yourself

 Worked example

Show that $1 \text{ km}^2 = 100 \text{ ha}$.

1. $1 \text{ km}^2 = 1000 \text{ m} \times 1000 \text{ m} = 1\,000\,000 \text{ m}^2$
2. $1 \text{ ha} = 10\,000 \text{ m}^2$
3. $1\,000\,000 \div 10\,000$

Answer: 100 ha

YOU DO

On your own: read the worked example, then try these.

- E1** How many mm^2 are in 1 m^2 ?

- E2** A soccer field is 105 m by 68 m. What fraction of a hectare is it (2 d.p.)?

Lesson summary

Missed the lesson? Start here.

Area conversions

1. $1 \text{ cm}^2 = 10 \times 10 = 100 \text{ mm}^2$.

2. $1 \text{ m}^2 = 100 \times 100 = 10\,000 \text{ cm}^2$.

3. $1 \text{ ha} = 10\,000 \text{ m}^2$; $1 \text{ km}^2 = 1\,000\,000 \text{ m}^2$.

4. To a smaller unit: $\times$. To a larger unit: $\div$.

 Level 1

9.1 m^2 to cm^2

1. $\times 10\,000$

Answer: 91 000 cm^2

 Level 2

240 mm^2 to cm^2

1. $\div 100$

Answer: 2.4 cm^2

 Level 3

240 000 mm^2 to m^2

1. $\div 1\,000\,000$

Answer: 0.24 m^2

 Extension

1 km^2 in ha

1. $1\,000\,000 \div 10\,000$

Answer: 100 ha

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LESSON
8.05

Area of squares and rectangles

Date _____

Learning intentions

We are learning to:

- find the area of squares and rectangles

Success criteria

I can:

- ☐ use $A = l \times w$ for a rectangle
- ☐ use $A = s^2$ for a square
- ☐ find a missing side from the area

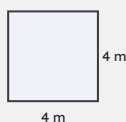
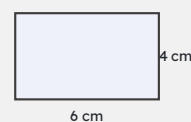
Key terms Write the definition as your teacher explains it.

Length

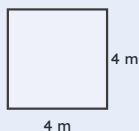
Width

Notes

Level 1 · Getting started

WE DO With your teacher: copy what they write on the board.**Example 1** Find the area.**Example 2** Find the area.**YOU DO** On your own: Set A. Find the area.

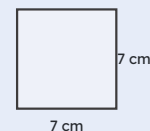
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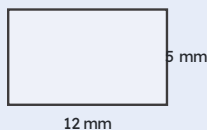
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2



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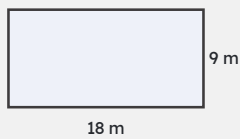


6



YOU DO On your own: Set B. Find the area.

- | | | | |
|---------------------------|----------------------|----------------------------|----------------------|
| 1 rectangle 15 cm by 8 cm | <input type="text"/> | 4 square with side 0.5 km | <input type="text"/> |
| 2 square with side 11 m | <input type="text"/> | 5 rectangle 40 mm by 25 mm | <input type="text"/> |
| 3 rectangle 3.5 m by 2 m | <input type="text"/> | 6 rectangle 1.2 m by 0.5 m | <input type="text"/> |

Level 2 · Building up**WE DO** With your teacher: copy what they write on the board.**Example 3** Find the area.

Example 4 A rectangle 2.5 m by 60 cm. Find its area in m^2 .
YOU DO On your own: Set C. Find the missing side.

- | | | |
|---|--|--|
| 1 A square has area 81 cm^2 . Find its side. | 3 A rectangle has area 7.5 cm^2 and width 2.5 cm. Find its length. | 5 A rectangle is 2.5 m by 60 cm. Find its area in m^2 . |
| <input type="text"/> | <input type="text"/> | <input type="text"/> |
| 2 A rectangle has area 48 m^2 and length 8 m. Find its width. | 4 A square has area 1.44 m^2 . Find its side. | 6 A square has perimeter 36 cm. Find its area. |
| <input type="text"/> | <input type="text"/> | <input type="text"/> |



Level 3 · Challenge

Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.**Example 5** A square has area 81 cm^2 . Find its side.
Example 6 A rectangle has area 48 m^2 and length 8 m. Find its width.
YOU DO On your own: Set D. Problems.

- 1 Carpet costs \$45 per m^2 . Find the cost to carpet a room 5.2 m by 4 m.

- 3 A rectangle has perimeter 26 cm and area 40 cm^2 . Find its length and width.

- 2 How many 30 cm by 30 cm tiles cover a floor 3.6 m by 2.7 m?

- 4 If you double the side of a square, what happens to its area?

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO

Show off your skill

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Find the area of a rectangle 9 cm by 6 cm.

Level 2

Find the area of a square with side 2.5 m.

Level 3A rectangle has area 60 m^2 and width 5 m. Find its length.

 Extension · Stretch yourself
 Worked example

A 20 m by 15 m lawn has a 2 m wide path around the outside of it. Find the area of the path.

1. Outer rectangle: 24 m by 19 m = 456 m²
2. Lawn: 20 × 15 = 300 m²
3. 456 – 300

Answer: 156 m²

YOU DO

On your own: read the worked example, then try these.

- E1** A 6 m by 4 m pool has a 1 m wide path around it. Find the path area.

- E2** Which rectangle with perimeter 24 cm has the largest area?

Lesson summary

Missed the lesson? Start here.

Remember 1. Area of a rectangle = length × width.
2. Use the same units for both sides.

2. Area of a square = side × side = s².
4. Answer in square units.

 Level 1

4 m square

1. 4 × 4

Answer: 16 m²

 Level 2

18 m × 9 m

1. 18 × 9

Answer: 162 m²

 Level 3

Area 48, length 8

1. 48 ÷ 8

Answer: 6 m

 Extension

Path around a lawn

1. 456 – 300

Answer: 156 m²

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LESSON
8.06

Area of a triangle

Date _____

Learning intentions

We are learning to:

- find the area of a triangle

Success criteria

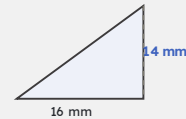
I can:

- ☐ identify the base and the perpendicular height
- ☐ use $A = \frac{1}{2} \times b \times h$
- ☐ find the height when it is outside the triangle

Key terms Write the definition as your teacher explains it.**Base****Perpendicular height**

Notes

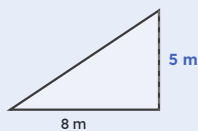
Level 1 · Getting started

WE DO With your teacher: copy what they write on the board.**Example 1** Find the area.**Example 2** Find the area.**YOU DO** On your own: Set A. Find the area.

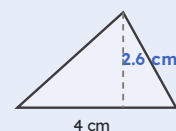
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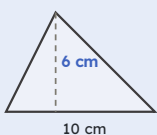
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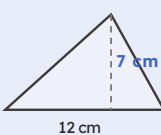
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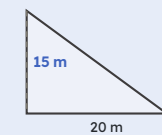
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4



6



YOU DO On your own: Set B. Find the area of each triangle.

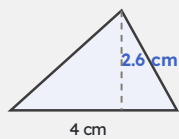
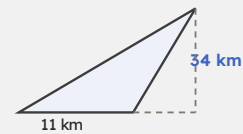
1 base 6 cm, height 4 cm

2 base 16 mm, height 14 mm

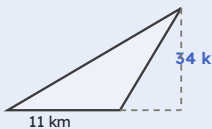
3 base 9 m, height 12 m

4 base 7 cm, height 5 cm

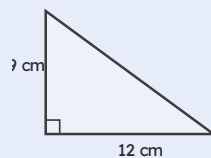
5 base 2.4 m, height 1.5 m

6 base 50 cm, height 1 m (in cm^2)**Level 2 · Building up****WE DO** With your teacher: copy what they write on the board.**Example 3** Find the area.**Example 4** Find the area.**YOU DO** On your own: Set C. Find the area. The height may be outside the triangle.

1



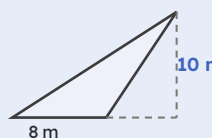
3



5

A triangle has area 20 m^2 and height 8 m. Find its base.

2



4

A triangle has area 36 cm^2 and base 9 cm. Find its height.

6

A right-angled triangle has shorter sides 5 cm and 12 cm. Find its area.



Level 3 · Challenge

Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.**Example 5** A triangle has area 36 cm^2 and base 9 cm. Find its height.
Example 6 Why is the area of a triangle half of a rectangle?
YOU DO On your own: Set D. Problems.

- 1 A triangular sail has base 3.2 m and height 5 m. Find its area.

- 3 Find the area of a square of side 10 cm with one corner triangle (legs 4 cm) cut off.

- 2 A triangular garden bed has base 6 m and height 4 m. Mulch covers 3 m^2 per bag. How many bags?

- 4 Two triangles have the same base 8 cm. One has height 5 cm, one 10 cm. Compare their areas.

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO Show off your skillChoose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Find the area: base 10 m, height 7 m.

Level 2

Find the area: base 5 cm, height 3.2 cm.

Level 3A triangle has area 24 cm^2 and base 6 cm. Find its height.

Extension · Stretch yourself

Worked example

Find the area of a kite with diagonals 10 cm and 16 cm.

1. The long diagonal splits it into 2 triangles, base 16, height 5
2. $2 \times \frac{1}{2} \times 16 \times 5$

Answer: 80 cm²

YOU DO

On your own: read the worked example, then try these.

- E1** Find the area of a rhombus with diagonals 6 m and 9 m.

- E2** A triangle has vertices (0, 0), (8, 0) and (3, 6) on a grid. Find its area.

Lesson summary

Missed the lesson? Start here.

Remember

1. $A = \frac{1}{2} \times \text{base} \times \text{height}$.
3. The height may be outside the triangle.

2. The height is perpendicular (at 90°) to the base.
4. Answer in square units.

Level 1

base 30, height 24

1. $\frac{1}{2} \times 30 \times 24$

Answer: 360 mm²

Level 2

base 4, height 2.6

1. $\frac{1}{2} \times 4 \times 2.6$

Answer: 5.2 cm²

Level 3

Area 36, base 9

1. $\frac{1}{2} \times 9 \times h = 36$

Answer: $h = 8$ cm

Extension

Kite 10 by 16

1. $2 \times \frac{1}{2} \times 16 \times 5$

Answer: 80 cm²

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
8.07

Area of a parallelogram

Date _____

Learning intentions

We are learning to:

- find the area of a parallelogram

Success criteria

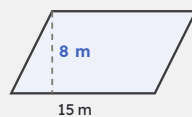
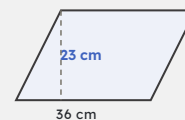
I can:

- ☐ identify the base and perpendicular height
- ☐ use $A = b \times h$
- ☐ not use the slanted side as the height

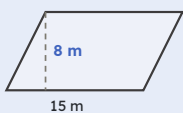
Key terms Write the definition as your teacher explains it.**Parallelogram****Perpendicular height**

Notes

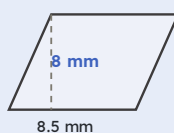
Level 1 · Getting started

WE DO With your teacher: copy what they write on the board.**Example 1** Find the area.**Example 2** Find the area.**YOU DO** On your own: Set A. Find the area.

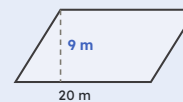
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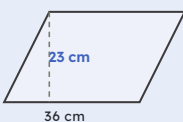
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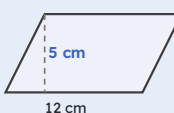
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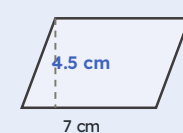
2



4



6



YOU DO On your own: Set B. Find the area of each parallelogram.

1 base 10 cm, height 6 cm

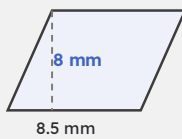
2 base 4.5 m, height 2 m

3 base 25 mm, height 12 mm

4 base 8 km, height 3.5 km

5 base 1.2 m, height 80 cm (m^2)

6 base 30 cm, height 15 cm

Level 2 · Building up**WE DO** With your teacher: copy what they write on the board.**Example 3** Find the area.**Example 4** Why is the area of a parallelogram the same as a rectangle with the same base and height?**YOU DO** On your own: Set C. Find the missing value.1 Area 72 cm^2 , base 12 cm. Find the height.3 Area 3.6 m^2 , base 1.8 m. Find the height.

5 Base 14 mm, slant side 9 mm, height 7 mm. Find the area.

2 Area 45 m^2 , height 5 m. Find the base.

4 Sides 10 cm and 6 cm, height 5 cm on the 10 cm side. Find the area.

6 A parallelogram and a triangle have base 8 cm and height 6 cm. Compare their areas.



Level 3 · Challenge

Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.**Example 5** A parallelogram has area 72 cm^2 and base 12 cm . Find its height.

Example 6 A parallelogram has sides 10 cm and 6 cm , and height 5 cm on the 10 cm side. Find its area.

YOU DO On your own: Set D. Problems.

- 1 A parallelogram-shaped garden has base 12 m and height 7.5 m . Turf costs $\$9.80$ per m^2 . Find the cost.

- 3 A rhombus has side 10 cm and height 8 cm . Find its area.

- 2 Draw two different parallelograms with area 24 cm^2 .

- 4 The same parallelogram has base 12 cm , height 6 cm , and another side 8 cm . Find the height on the 8 cm side.

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO

Show off your skill

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1Find the area: base 9 cm , height 4 cm .

Level 2Find the area: base 2.5 m , height 1.6 m .

Level 3A parallelogram has area 50 m^2 and height 4 m . Find its base.

 Extension · Stretch yourself

 Worked example

Find the area of a trapezium with parallel sides 8 cm and 14 cm and height 5 cm.

- Two copies make a parallelogram with base $8 + 14 = 22$
- Half of 22×5

Answer: 55 cm^2

YOU DO

On your own: read the worked example, then try these.

- E1** Find the area of a trapezium with parallel sides 6 m and 10 m and height 4 m.

- E2** A trapezium has area 60 cm^2 , height 6 cm and one parallel side 8 cm. Find the other.

Lesson summary

Missed the lesson? Start here.

Remember

- $A = \text{base} \times \text{height}$.
- Height is perpendicular to the base.
- Do not use the slanted side.
- Cut the triangle off one end and move it: a rectangle.

 Level 1

base 15, height 8

1. 15×8

Answer: 120 m^2

 Level 2

base 8.5, height 8

1. 8.5×8

Answer: 68 mm^2

 Level 3

Area 72, base 12

1. $72 \div 12$

Answer: 6 cm

 Extension

Trapezium 8, 14, h 5

1. $\frac{1}{2} \times 22 \times 5$

Answer: 55 cm^2

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
8.08

Area of composite shapes

Date _____

Learning intentions

We are learning to:

- find the area of composite and shaded shapes

Success criteria

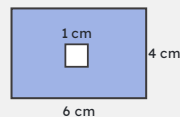
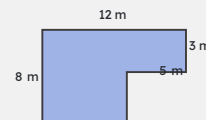
I can:

- ☐ split a shape into rectangles and triangles, then add
- ☐ find a shaded area by subtracting
- ☐ solve practical problems such as painting a wall

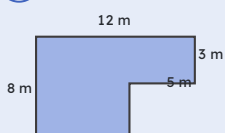
Key terms Write the definition as your teacher explains it.**Composite shape****Shaded area**

Notes

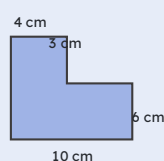
Level 1 · Getting started

WE DO With your teacher: copy what they write on the board.**Example 1** Find the shaded area.**Example 2** Find the area.**YOU DO** On your own: Set A. Find the area by splitting into rectangles.

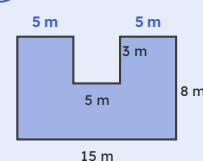
1



2

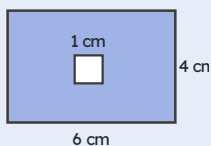


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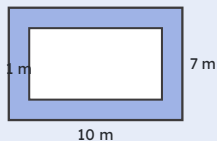


YOU DO On your own: Set B. Find the shaded area.

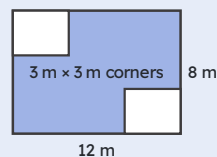
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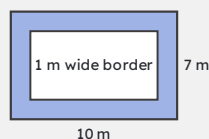
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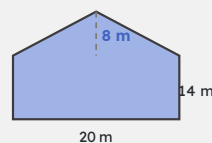
Level 2 · Building up

WE DO With your teacher: copy what they write on the board.

Example 3 Find the shaded area.

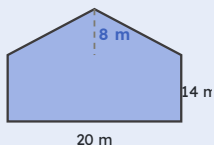


Example 4 Find the area.

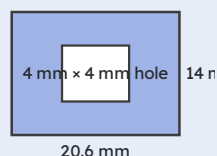


YOU DO On your own: Set C. Find the area.

1

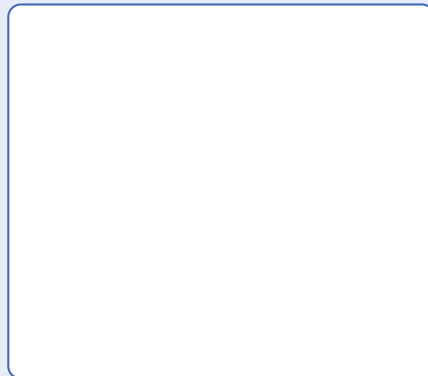


3



5

A 10 m by 6 m rectangle with a triangle of base 10 m and height 4 m on top.

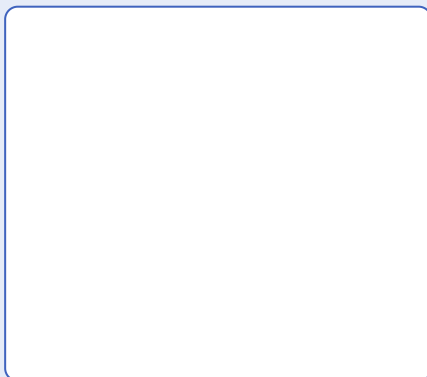


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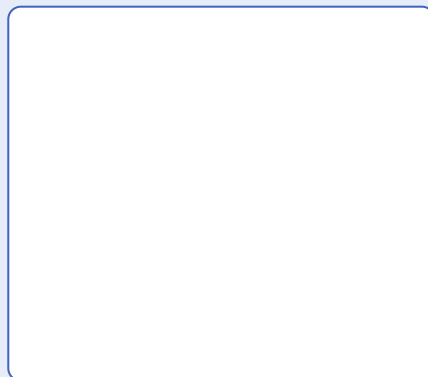
4

A rectangle 12 cm by 8 cm with a 4 cm by 3 cm rectangle cut from one corner.



6

A square of side 8 cm with a square of side 3 cm cut from the centre.





Level 3 · Challenge

Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.

Example 5 A wall 6.6 m by 3 m has a 2 m by 1 m window and a 0.9 m by 2 m door. One litre of paint covers 3 m^2 . How many whole litres are needed?

Example 6 Find the area of a square of side 5 mm with a triangle of base 5 mm and height 5 mm on top.

YOU DO On your own: Set D. Problems.

- 1 A wall 6.6 m by 3 m has a 2 m by 1 m window and a 0.9 m by 2 m door. One litre of paint covers 3 m^2 . How many whole litres are needed?

- 3 Find the area of an L-shaped room: 6 m by 4 m with a 2 m by 1.5 m corner cut out.

- 2 A 50 cm by 35 cm poster has a 40 cm by 28 cm photo on it. Find the area of the poster not covered.

- 4 Carpet costs \$38 per m^2 . Find the cost of carpeting the room in Question 3.

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO Show off your skill

Choose ONE question. Show your working.

Name _____ Class _____

Level 1

Find the area of a $10 \text{ cm} \times 5 \text{ cm}$ rectangle with a $2 \text{ cm} \times 2 \text{ cm}$ square hole.

Level 2

Find the area of an L-shape made from a $6 \text{ m} \times 3 \text{ m}$ and a $2 \text{ m} \times 2 \text{ m}$ rectangle.

Level 3

A wall is $4 \text{ m} \times 2.5 \text{ m}$ with a $1 \text{ m} \times 1 \text{ m}$ window. Paint covers 3 m^2 per litre. How many whole litres?

Extension · Stretch yourself

Worked example

A path 1 m wide runs along two sides of a 12 m by 8 m lawn (an L-shape). Find the area of the path.

1. Outer rectangle $13 \times 9 = 117 \text{ m}^2$
2. Lawn $12 \times 8 = 96 \text{ m}^2$
3. $117 - 96$

Answer: 21 m^2

YOU DO

On your own: read the worked example, then try these.

- E1** A 1.5 m wide path goes around all four sides of a 10 m by 6 m pool. Find its area.

- E2** Find the area of a cross made of five 3 cm squares.

Lesson summary

Missed the lesson? Start here.

- Remember**
1. Split the shape into rectangles and triangles.
 2. Find missing lengths first.
 3. Add the parts, or subtract holes (shaded area).
 4. Check units and give square units.

Level 1

6×4 rectangle with 1×1 hole

1. $24 - 1$

Answer: 23 cm^2

Level 2

L-shape 12×8 minus 5×5

1. $96 - 25$

Answer: 71 m^2

Level 3

Border 1 m wide on 10×7

1. $70 - 8 \times 5$

Answer: 30 m^2

Extension

Paint the wall

1. $19.8 - 2 - 1.8 = 16 \text{ m}^2$
2. $16 \div 3 = 5.3$

Answer: 6 L

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LESSON
8.09

Prisms, cross-sections and views

Date _____

Learning intentions

We are learning to:

- identify prisms and their cross-sections
- draw the front, side and top views of a solid

Success criteria

I can:

- ☐ know a prism has the same cross-section all the way through
- ☐ draw a cross-section parallel to the ends
- ☐ sketch front, side and top views

Key terms Write the definition as your teacher explains it.

Prism

Cross-section

View

Notes

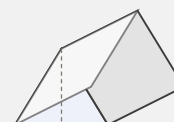
Level 1 · Getting started

WE DO With your teacher: copy what they write on the board.

Example 1 Is it a prism? Draw its cross-section.

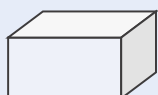


Example 2 Is it a prism? Draw its cross-section.

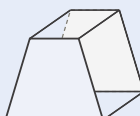


YOU DO On your own: Set A. Is each solid a prism? If so, draw its cross-section.

1



3

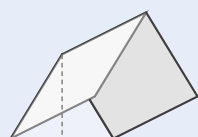


5

a square pyramid



2

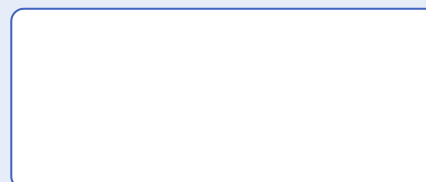


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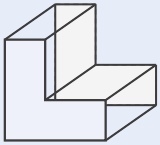
6

a cylinder

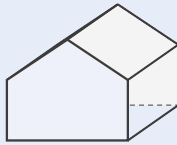


YOU DO On your own: Set B. Draw the cross-section of each prism.

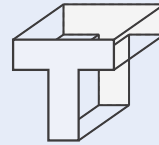
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2



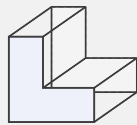
3



Level 2 · Building up

WE DO With your teacher: copy what they write on the board.

Example 3 Draw the cross-section.

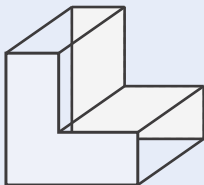


Example 4 Is a cone a prism? Explain.



YOU DO On your own: Set C. Sketch the front (F), side and top views.

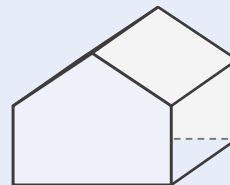
1 front, left and top views



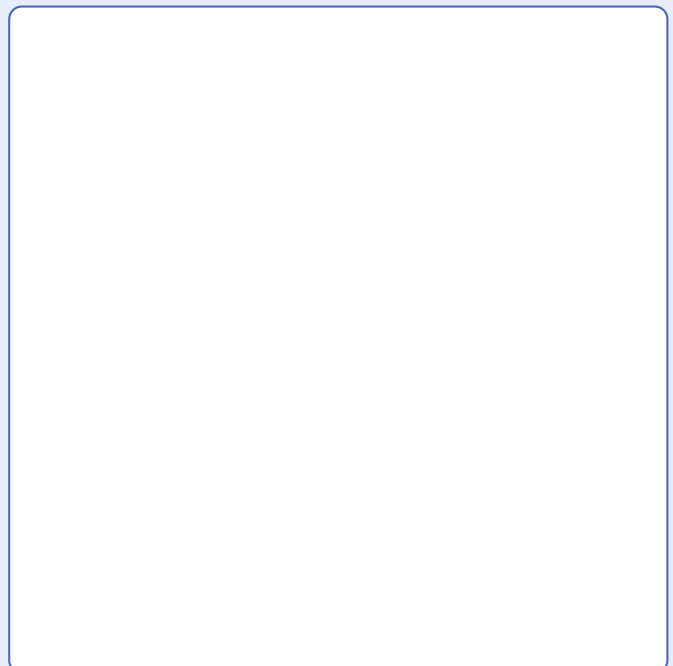
F



2 right, front and top views



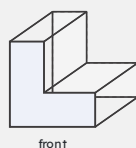
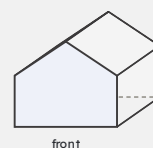
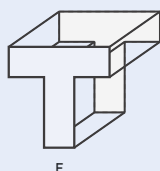
F





Level 3 · Challenge

Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.**Example 5** Sketch the front, left and top views.**Example 6** Sketch the right, front and top views.**YOU DO** On your own: Set D. Views and solids.**1** Sketch the front, side and top views.**3** Name a solid whose front, side and top views are all the same.
2 A solid has a square front view, a square side view and a circle top view. What is it?
4 Draw the top view of a cylinder lying on its side.

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO

Show off your skill

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Is a cylinder a prism? Explain.

Level 2

Name the cross-section of a hexagonal prism.

Level 3

Sketch the top view of a triangular prism lying on its rectangular face.

Extension · Stretch yourself

Worked example

A solid has front view a rectangle, side view a triangle and top view a rectangle. What is it?

1. A triangle from the side means triangular ends
2. Rectangles from the front and top: it is lying on a rectangular face

Answer: a triangular prism

YOU DO

On your own: read the worked example, then try these.

- E1** A solid has front and side views both triangles and top view a square. What is it?

- E2** Draw the front view of a stack of cubes: 3 in the back row, 2 in the middle, 1 at the front, all in a line.

Lesson summary

Missed the lesson? Start here.

- Remember**
1. A **prism** has two identical, parallel ends joined by rectangles.
 2. The **cross-section** is the same shape all the way through.
 3. Views: look from the front, the side and the top.
 4. Pyramids, cones and spheres are not prisms.

Level 1

Rectangular box

1. Same rectangle all through

Answer: prism, rectangle cross-section

Level 2

Cone

1. Cross-section shrinks

Answer: not a prism

Level 3

L-shaped prism

1. Front: an L

Answer: L-shape, rectangle, rectangle

Extension

Rectangle, triangle, rectangle

1. Triangular ends

Answer: triangular prism

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LESSON
8.10

Volume units

Date _____

Learning intentions

We are learning to:

- convert between units of volume

Success criteria

I can:

- ☐ know $1 \text{ cm}^3 = 1000 \text{ mm}^3$ and $1 \text{ m}^3 = 1\,000\,000 \text{ cm}^3$
- ☐ multiply to a smaller unit, divide to a larger unit
- ☐ count cubes to find volume

Key terms Write the definition as your teacher explains it.**Volume****Cubic unit**

Notes

Level 1 · Getting started

WE DO With your teacher: copy what they write on the board.**Example 1** Why is $1 \text{ cm}^3 = 1000 \text{ mm}^3$?**Example 2** $20 \text{ cm}^3 = ___ \text{ mm}^3$ **YOU DO** On your own: Set A. Convert.1 20 cm^3 to mm^3

4 0.5 m^3 to cm^3

2 3 cm^3 to mm^3

5 4.5 cm^3 to mm^3

3 2 m^3 to cm^3

6 1.2 m^3 to cm^3

YOU DO On your own: Set B. Convert.

1 7500 mm^3 to cm^3

4 $250\,000 \text{ cm}^3$ to m^3

2 4000 mm^3 to cm^3

5 600 mm^3 to cm^3

3 $3\,000\,000 \text{ cm}^3$ to m^3

6 $80\,000 \text{ cm}^3$ to m^3

Level 2 · Building up**WE DO** With your teacher: copy what they write on the board.

Example 3 $0.5 \text{ m}^3 = \text{---} \text{ cm}^3$

Example 4 $7500 \text{ mm}^3 = \text{---} \text{ cm}^3$

YOU DO On your own: Set C. Convert. Show your working.

1 $230\,000 \text{ mm}^3$ to m^3

3 5 km^3 to m^3

5 $12\,500 \text{ cm}^3$ to m^3

2 0.004 m^3 to mm^3

4 0.75 m^3 to cm^3

6 9 cm^3 to m^3



Level 3 · Challenge

Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.**Example 5** $230\,000\text{ mm}^3 = ___\text{ m}^3$

Example 6 How many 1 cm cubes fit in a 5 cm by 4 cm by 3 cm box?
YOU DO On your own: Set D. Cubes and volume.**1** How many 1 cm cubes fit in a box 5 cm by 4 cm by 3 cm?
3 Order: 0.02 m^3 , $15\,000\text{ cm}^3$, $25\,000\,000\text{ mm}^3$.
2 How many 2 cm cubes fit in a box 10 cm by 6 cm by 4 cm?
4 Which unit for: a sugar cube, a swimming pool, a lunch box?

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO Show off your skillChoose **ONE** question. Show your working.

Name _____ Class _____

Level 1Convert 6 cm^3 to mm^3 .
Level 2Convert $2\,500\,000\text{ cm}^3$ to m^3 .
Level 3Convert 1500 mm^3 to cm^3 .

Extension · Stretch yourself

Worked example

How many mm^3 in 1 m^3 ?

1. $1 \text{ m} = 1000 \text{ mm}$
2. $1000 \times 1000 \times 1000$

Answer: $1\,000\,000\,000 \text{ mm}^3$

YOU DO

On your own: read the worked example, then try these.

- E1** Convert $0.000\,35 \text{ m}^3$ to mm^3 .

- E2** A cube has volume 1 m^3 . How many 10 cm cubes fit inside it?

Lesson summary

Missed the lesson? Start here.

Volume conversions

1. $1 \text{ cm}^3 = 10 \times 10 \times 10 = 1000 \text{ mm}^3$.

2. $1 \text{ m}^3 = 100 \times 100 \times 100 = 1\,000\,000 \text{ cm}^3$.

3. To a smaller unit: $\times$. To a larger unit: $\div$.

4. Volume is measured in cubic units.

Level 1

20 cm^3 to mm^3

1. $\times 1000$

Answer: $20\,000 \text{ mm}^3$

Level 2

0.5 m^3 to cm^3

1. $\times 1\,000\,000$

Answer: $500\,000 \text{ cm}^3$

Level 3

$230\,000 \text{ mm}^3$ to m^3

1. $\div 1\,000\,000\,000$

Answer: 0.00023 m^3

Extension

1 m^3 in mm^3

1. 1000^3

Answer: $1\,000\,000\,000$

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LESSON
8.11

Volume of a rectangular prism

Date _____

Learning intentions

We are learning to:

- find the volume of rectangular prisms

Success criteria

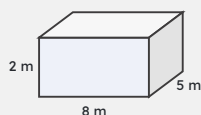
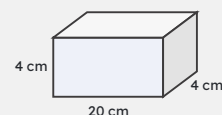
I can:

- ☐ use $V = l \times w \times h$
- ☐ use $V = \text{area of base} \times \text{height}$
- ☐ find a missing dimension

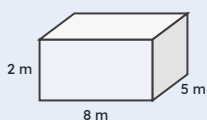
Key terms Write the definition as your teacher explains it.**Volume****Base area**

Notes

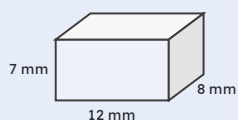
Level 1 · Getting started

WE DO With your teacher: copy what they write on the board.**Example 1** Find the volume.**Example 2** Find the volume.**YOU DO** On your own: Set A. Find the volume.

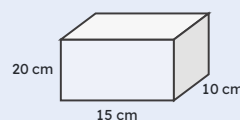
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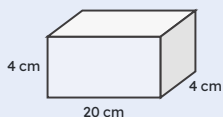
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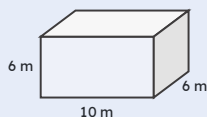
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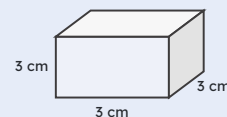
2



4



6



YOU DO On your own: Set B. Find the volume.

1 $6 \text{ cm} \times 5 \text{ cm} \times 2 \text{ cm}$

2 cube with edge 5 m

3 $2.5 \text{ m} \times 2 \text{ m} \times 1 \text{ m}$

4 $10 \text{ mm} \times 10 \text{ mm} \times 3 \text{ mm}$

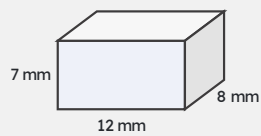
5 base area 24 cm^2 ,
height 5 cm

6 $1.2 \text{ m} \times 0.5 \text{ m} \times 0.5 \text{ m}$

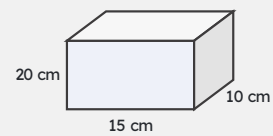
Level 2 · Building up

WE DO With your teacher: copy what they write on the board.

Example 3 Find the volume.



Example 4 Find the volume.



YOU DO On your own: Set C. Find the missing value.

1 A cube has volume 64 cm^3 . Find its edge.

3 Volume 90 m^3 , base area 15 m^2 . Find the height.

5 A box is 40 cm by 30 cm by 25 cm. Find its volume in m^3 .

2 Volume 120 cm^3 , length 6 cm, width 4 cm. Find the height.

4 A cube has volume 1000 mm^3 . Find its edge.

6 Double every edge of a 2 cm cube. How many times bigger is the volume?



Level 3 · Challenge

Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.**Example 5** A cube has volume 64 cm^3 . Find its edge length.
Example 6 A box has volume 120 cm^3 , length 6 cm and width 4 cm. Find its height.
YOU DO On your own: Set D. Problems.**1** A shipping container is 12 m by 2.4 m by 2.6 m. Find its volume.
3 Sand is piled into a trailer 2 m by 1.2 m by 0.4 m. Find the volume of sand.
2 How many 5 cm cubes fit in a box 30 cm by 20 cm by 15 cm?
4 A rectangular prism has volume 48 cm^3 . Find 3 possible sets of whole-number dimensions.

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO Show off your skillChoose **ONE** question. Show your working.

Name _____ Class _____

Level 1Find the volume: $5 \text{ cm} \times 4 \text{ cm} \times 3 \text{ cm}$.
Level 2

Find the volume of a cube with edge 6 m.

Level 3A box has volume 72 cm^3 and base area 12 cm^2 . Find its height.

 Extension · Stretch yourself
 Worked example

An L-shaped prism has a cross-section of area 90 m^2 and length 7 m. Find its volume.

1. $V = \text{area of cross-section} \times \text{length}$
2. 90×7

Answer: 630 m^3

YOU DO

On your own: read the worked example, then try these.

- E1** A prism has an L-shaped cross-section of area 45 cm^2 and length 12 cm. Find its volume.

- E2** A room is 5 m by 4 m by 2.7 m. An air conditioner cycles 1 m^3 of air every 2 seconds. How long to cycle all the air?

Lesson summary

Missed the lesson? Start here.

Remember 1. $V = \text{length} \times \text{width} \times \text{height}$.
3. Cube: $V = s^3$.

2. Or $V = \text{area of base} \times \text{height}$.
4. Answer in cubic units.

 Level 1

$$8 \times 5 \times 2$$

1. 40×2

Answer: 80 m^3

 Level 2

$$15 \times 10 \times 20$$

1. 150×20

Answer: 3000 cm^3

 Level 3

$$\text{Cube } 64 \text{ cm}^3$$

1. $4 \times 4 \times 4 = 64$

Answer: 4 cm

 Extension

L-prism: area 90, length 7

1. 90×7

Answer: 630 m^3

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
8.12

Volume of a prism

Date _____

Learning intentions

We are learning to:

- find the volume of prisms with triangular and other cross-sections

Success criteria

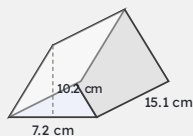
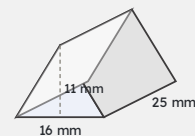
I can:

- ☐ find the area of the cross-section first
- ☐ use $V = A \times h$ (h is the length of the prism)
- ☐ use this for triangular, trapezium and composite prisms

Key terms Write the definition as your teacher explains it.**Cross-section****Prism**

Notes

Level 1 · Getting started

WE DO With your teacher: copy what they write on the board.**Example 1** Find the volume.**Example 2** Find the volume.**YOU DO** On your own: Set A. Find the volume. (A = cross-section area, h = length)

1 $A = 20 \text{ cm}^2$, $h = 5 \text{ cm}$

4 $A = 7.5 \text{ mm}^2$, $h = 4 \text{ mm}$

2 $A = 35 \text{ cm}^2$, $h = 8 \text{ cm}$

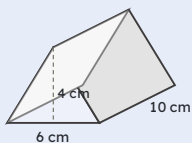
5 $A = 100 \text{ cm}^2$, $h = 0.5 \text{ cm}$

3 $A = 12 \text{ m}^2$, $h = 3 \text{ m}$

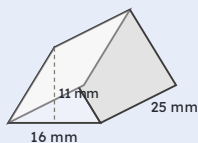
6 $A = 48 \text{ m}^2$, $h = 10 \text{ m}$

YOU DO On your own: Set B. Find the area of the triangular cross-section, then the volume.

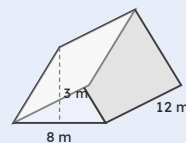
1



2



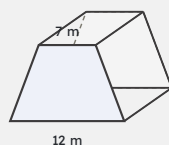
3



Level 2 · Building up

WE DO With your teacher: copy what they write on the board.

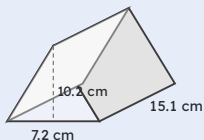
Example 3 Find the volume (trapezium cross-section: parallel sides 7 m and 12 m, height 6 m, length 10 m).



Example 4 A prism has cross-section area 35 cm^2 and length 8 cm. Find its volume.

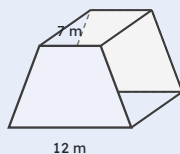
YOU DO On your own: Set C. Find the volume.

1



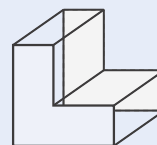
2

trapezium: 7 m and 12 m, height 6 m; length 10 m



3

L-shape area 45 cm^2 , length 12 cm





Level 3 · Challenge

Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.**Example 5** Find the volume of a triangular prism: triangle base 6 cm, height 4 cm, prism length 10 cm.
Example 6 A prism has volume 360 m^3 and length 12 m. Find the area of its cross-section.
YOU DO On your own: Set D. Problems.

- 1 A tent is a triangular prism: triangle base 2.4 m, height 1.5 m, length 3 m. Find its volume.

- 3 A prism has volume 360 m^3 and length 12 m. Find the cross-section area.

- 2 A ramp has a right-triangle cross-section with legs 1.2 m and 0.5 m and width 2 m. Find its volume.

- 4 A water trough has a trapezium cross-section (parallel sides 60 cm and 40 cm, depth 30 cm) and length 2 m. Find its volume in cm^3 .

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO

Show off your skill

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1A prism has cross-section 15 cm^2 and length 6 cm. Find its volume.
Level 2

Triangle base 5 m, height 4 m, prism length 9 m. Find the volume.

Level 3A prism has volume 200 cm^3 and cross-section 25 cm^2 . Find its length.

Extension · Stretch yourself

Worked example

A swimming pool is 25 m long and 10 m wide. It is 1 m deep at one end and 2 m deep at the other. Find its volume.

1. Cross-section (side view) is a trapezium: $\frac{1}{2} \times (1 + 2) \times 25 = 37.5 \text{ m}^2$
2. Width 10 m: 37.5×10

Answer: 375 m^3

YOU DO

On your own: read the worked example, then try these.

- E1** A pool 50 m by 20 m is 1.2 m deep at one end and 2 m at the other. Find its volume.

- E2** How many litres is the pool in the worked example? ($1 \text{ m}^3 = 1000 \text{ L}$)

Lesson summary

Missed the lesson? Start here.

Remember

1. Identify the cross-section (the end face).
2. Find its area A .
3. $V = A \times \text{length of the prism}$.
4. Triangle: $A = \frac{1}{2}bh$. Trapezium: $A = \frac{1}{2}(a + b)h$.

Level 1

Triangle 7.2×10.2 ,
length 15.1

1. $A = \frac{1}{2} \times 7.2 \times 10.2 = 36.72$
2. $\times 15.1$

Answer: 554.472 cm^3

Level 2

$A = 35$, $h = 8$

1. 35×8

Answer: 280 cm^3

Level 3

Trapezium 7, 12, height
6, length 10

1. $\frac{1}{2} \times 19 \times 6 = 57$
2. $\times 10$

Answer: 570 m^3

Extension

Sloping pool

1. 37.5×10

Answer: 375 m^3

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
8.13

Capacity

Date _____

Learning intentions

We are learning to:

- find the capacity of a container in litres

Success criteria

I can:

- ☐ know $1 \text{ cm}^3 = 1 \text{ mL}$ and $1 \text{ m}^3 = 1000 \text{ L}$
- ☐ find volume, then convert to litres
- ☐ solve problems about water levels and large volumes

Key terms Write the definition as your teacher explains it.**Capacity****Megalitre**

Notes

Level 1 · Getting started

WE DO With your teacher: copy what they write on the board.**Example 1** A box is 30 cm by 20 cm by 10 cm. Find its capacity in litres.
Example 2 The water is poured into a prism with base 20 cm by 20 cm. How high does it reach?
YOU DO On your own: Set A. Convert.1 500 cm^3 to mL
2 2000 cm^3 to L
3 3.5 m^3 to L
4 0.2 m^3 to L

5 4500 L to kL

6 2 ML to L

YOU DO On your own: Set B. Find the capacity in litres.

1 $30\text{ cm} \times 20\text{ cm} \times 10\text{ cm}$

4 $50\text{ cm} \times 40\text{ cm} \times 30\text{ cm}$

2 $8\text{ m} \times 5\text{ m} \times 2\text{ m}$

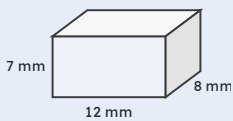
5 $1\text{ m} \times 1\text{ m} \times 1\text{ m}$

3 $20\text{ cm} \times 4\text{ cm} \times 4\text{ cm}$

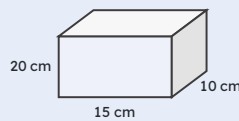
6 $10\text{ cm} \times 10\text{ cm} \times 10\text{ cm}$

Level 2 · Building up**WE DO** With your teacher: copy what they write on the board.**Example 3** Convert 3.5 m^3 to litres.**Example 4** Find the capacity in litres of a tank 8 m by 5 m by 2 m.**YOU DO** On your own: Set C. Find the capacity in litres of each prism.

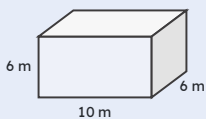
1



3

5 trapezium prism: 570 m^3

2



4

triangular prism: $\frac{1}{2} \times 16 \times 11$ mm, length 25 mm

6 cube, edge 2.5 m



Level 3 · Challenge

Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.**Example 5** An iceberg is like a prism 160 km by 50 km by 250 m. How many megalitres of water?
Example 6 A 2 L jug fills cups of 250 mL. How many cups?
YOU DO On your own: Set D. Problems.

- 1 A 30 cm × 20 cm × 10 cm box of water is poured into a prism with base 20 cm × 20 cm. How high does the water reach?

- 3 A fish tank 60 cm × 30 cm × 40 cm is filled to 5 cm from the top. How many litres of water?

- 2 Iceberg B9 was like a prism 160 km by 50 km by 250 m. How many megalitres of water did it hold?

- 4 A 1.5 m by 0.8 m bath fills at 12 L a minute. How deep is the water after 10 minutes (cm)?

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO Show off your skillChoose **ONE** question. Show your working.

Name _____ Class _____

Level 1Convert 2500 cm³ to litres.
Level 2

Find the capacity in litres of a box 25 cm × 20 cm × 10 cm.

Level 3Convert 0.6 m³ to litres.

 Extension · Stretch yourself

 Worked example

Rain of 10 mm falls on a 12 m by 8 m roof. How many litres run into the tank?

1. 10 mm = 0.01 m
2. $V = 12 \times 8 \times 0.01 = 0.96 \text{ m}^3$
3. $\times 1000$

Answer: 960 L

YOU DO

On your own: read the worked example, then try these.

- E1** How much rain (mm) must fall on the roof to fill a 4800 L tank?

- E2** A house uses 180 L per person per day. How many days does 960 L last a family of 4?

Lesson summary

Missed the lesson? Start here.

Capacity 1. $1 \text{ cm}^3 = 1 \text{ mL}$; $1000 \text{ cm}^3 = 1 \text{ L}$.
3. 1 ML (megalitre) = 1 000 000 L.

2. $1 \text{ m}^3 = 1000 \text{ L} = 1 \text{ kL}$.
4. Find the volume first, then convert.

 Level 1

$30 \times 20 \times 10 \text{ cm}$

1. 6000 cm^3

Answer: 6 L

 Level 2

Pour into 20×20 base

1. $6000 \div 400$

Answer: 15 cm (D)

 Level 3

$8 \times 5 \times 2 \text{ m}$

1. $80 \text{ m}^3 \times 1000$

Answer: 80 000 L

 Extension

Rain on a roof

1. 0.96 m^3

Answer: 960 L

This space is left blank on purpose. It is the back of your exit ticket.